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RESEARCH ARTICLE

Spatiotemporal patterns and evolution of heavy rainfall trajectories in China

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Abstract

A heavy rainfall trajectory (HRT) is defined as the trajectory formed by the movement of the rainstorm centre of a heavy rainfall event over time. Based on this definition, the dispersion degree (DD) of HRTs is applied based on the Euclidean distance and an empirical orthogonal function (EOF) to quantify the spatiotemporal distribution and evolution characteristics of HRTs in China. The results show that (a) the spatial distribution and length of HRTs in China have distinct geographical and seasonal characteristics, which are strongly influenced by the East Asian summer monsoon. (b) From southeast China to northwest China, the DDs increased to 8 and the length of HRTs decreased by 53,925 m. (c) According to EOF analysis, the direction angle change of northeast-oriented HRTs shows a significant negative trend across the country, while those of southeast-, southwest- and northwest-oriented HRTs show a significant positive trend. From 2008 to 2019, HRTs flattened in the northeast and southwest directions, while HRTs steepened in the southeast and northwest directions. This research provides certain scientific guidance for the prevention and mitigation of regional disasters and provides necessary theoretical support for the changing patterns of the climate system under the impact of climate change.

KEYWORDS

China, dispersion, EOF analysis, heavy rainfall trajectory, spatiotemporal patterns

1 | INTRODUCTION

The changing patterns of heavy rainfall events, which are associated with both thermodynamical processes and dynamical processes (Moore *et al.*, 2015; Fu *et al.*, 2016; Kumari *et al.*, 2021), cause an increase in heavy rainfall. The former is easier to quantify than the latter, whose rate closely follows the Clausius–Clapeyron relationship on a global scale (IPCC, 2021). According to the data from Global Disaster Data Platform, in the past decade, 171 heavy rainfall events have occurred worldwide, which caused 16,837 deaths and 654,592.4 million dollars in losses. The disasters (floods, landslides, tropical storms, etc.) caused by heavy rainfall events have a ratio greater than 80% of the total global disasters. Therefore, it is necessary to further learn the spatiotemporal distribution and evolution of heavy rainfall events, which has significant meaning for the early warning of regional heavy rainfall events.

Heavy rainfall events on daily and hourly scales always cause severe flood disasters and substantial losses in life and property. Researchers are focusing on forecasting rainfall using attribution analysis, which is an accurate method for predicting future precipitation. Many meteorological systems, such as tropical cyclones and mechanisms, have been shown to significantly contribute

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TABLE 1 The main occurrence time and causes of heavy rainfall events in the eight geographical subregions of China

Region of China	Main occurrence time (Luo *et al.*, 2020)	Main causes (Chang, 2008; Luo *et al.*, 2020)	Transport and sources of water vapour (Luo *et al.*, 2020; Sun *et al.*, 2020)
Northeast China	The strongest stage of East Asian summer monsoons (Jul–Aug)	Northeast China cold eddy	Moisture channel of mid-latitude westerly belt. Southwest moisture channel of Indian monsoons. Moisture channel in the western side of western Pacific subtropical high
North China	After Meiyu period on the Changjiang-Huaihe River	Low-pressure trough and vortex rainstorm (Gao *et al.*, 2018), shear line, cold front with trough (Kang *et al.*, 2020)	Moisture channel of mid-latitude westerly belt. Southwest moisture channel of Indian monsoons. Moisture channel in the western side of western Pacific subtropical high
Middle and lower reaches area of Changjiang-Huaihe River	Meiyu period on the Changjiang-Huaihe River (mid to late Jun to early Jul)	Meiyu front	Southwest moisture channel of Indian monsoons. Moisture channel in the western side of western Pacific subtropical high
South China	Apr–Sep	The preflood season (Apr–Jun) is influenced by the East Asian subtropical monsoon and westerly systems. The after-flood season (Jul–Sep) is influenced by the tropical weather systems	Southwest moisture channel of Indian monsoons. Moisture channel in the western side of western Pacific subtropical high
Southwest China	May–Aug	Southwest China vortex	Southwest moisture channel of Indian monsoons
Tibetan Plateau region	May–Aug	Low-pressure trough, low vortex and shear line (Zhao, 2012)	Moisture channel of mid-latitude westerly belt. Southwest moisture channel of Indian monsoons
Northwest China	Jul–Aug	Westerly belt, subtropical and tropical circulation systems (Huang *et al.*, 2019)	Moisture channel of mid-latitude westerly belt

to heavy or extreme precipitation (Barlow *et al.*, 2019), particularly synoptic systems (Rabinowitz *et al.*, 2019; Wu and Luo, 2019). Many physics-based models of different synoptic scales, such as the Weather Research and Forecasting (WRF) Model, Regional Climate Model (RCM) and Global Coupled Models (GCMs), have been employed to predict future climate change and to assess the impact of climate change on precipitation (Peltier *et al.*, 2018; Ren *et al.*, 2020). In addition, the triggering and amplifying function of mountains for mesoscale convective systems (Alexander *et al.*, 2015; Eghdami and Barros, 2019) can impact heavy rainfall events regarding their magnitude of extreme precipitation events, the number of days (Lemus-Canovas *et al.*, 2021) and the intensity of daily and subdaily extremes (Panziera *et al.*, 2018; Kang *et al.*, 2020). Current heavy rainfall studies almost always use indicators such as the amount and intensity of heavy rainfall and the number of heavy rainfall days to quantify the degree and trend of heavy rainfall spatiotemporal distribution and evolution (Lin and Yang, 2014; Ren *et al.*, 2015; Kong *et al.*, 2017; He and Zhai, 2018). These studies disregard spatial patterns such as the direction of the same heavy rainfall within the life cycle and hinder further research on the spatiotemporal evolution of persistent heavy rainfall. These two factors have important theoretical and application values for meteorological disaster prevention and climate change response.

In China, heavy rainfall events are prevalent, and climate change has increased the frequency and intensity of heavy rainfall events (Ren *et al.*, 2015; Tang *et al.*, 2019). The average increasing trend of maximum hourly rainfall intensity in China during the summer is approximately 11.2% (Xiao *et al.*, 2016). The changing trend of heavy rainfall events is dominated by high spatial variability, and the persistence of heavy rainfall is a distinctive feature of heavy rainfall in China (Ding, 2019). Different data types and study areas lead to various definitions for persistent heavy rainfall events on a daily scale (Lin *et al.*, 2020). Persistent heavy rainfall events are almost always defined as heavy rainfall events with a minimum period of 3 days (Chen, 2013). Most daily scale studies of persistent heavy rainfall events emphasize the persistence of the occurrence time in the study area while disregarding heavy rainfall

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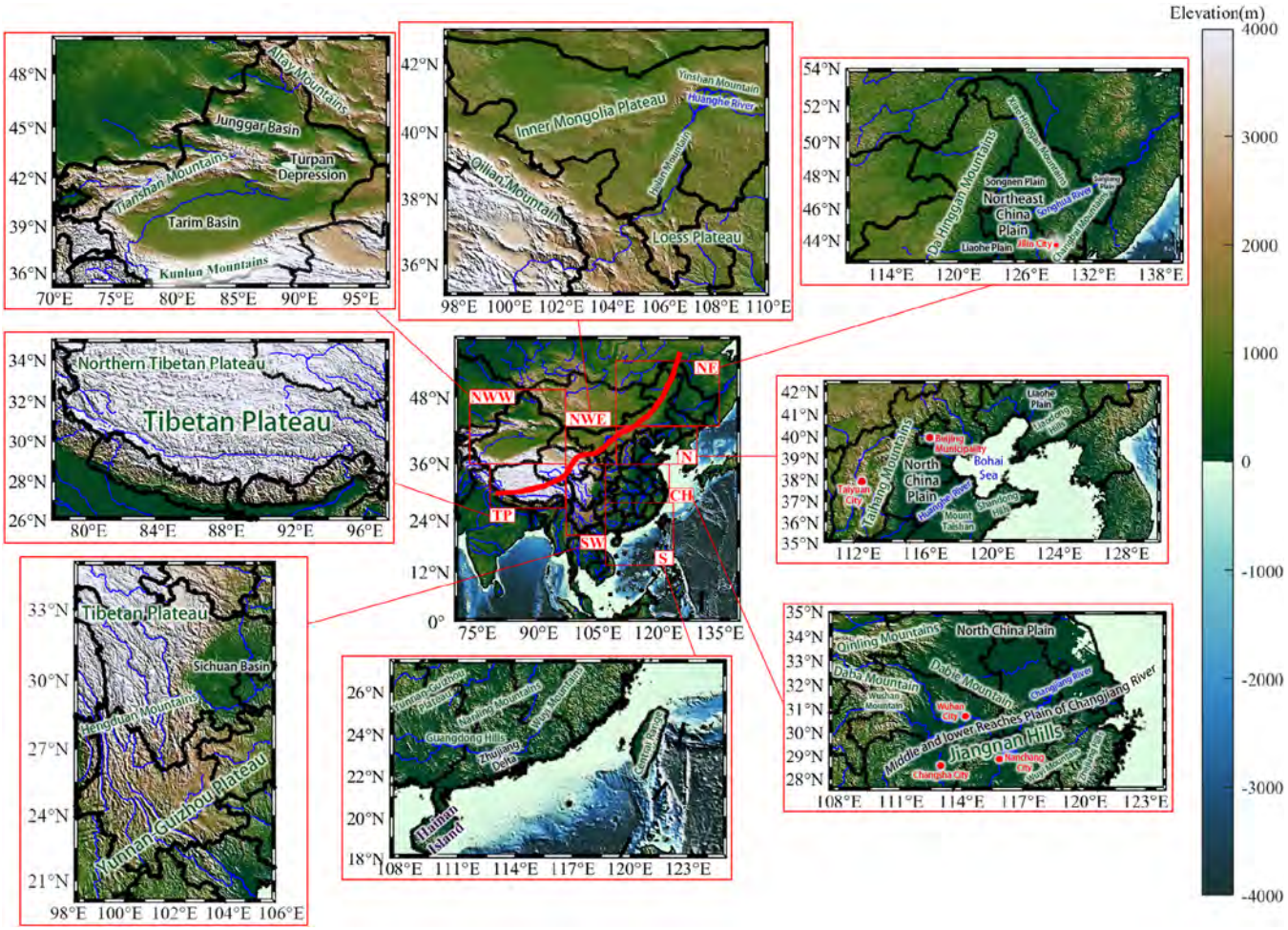


FIGURE 1 Geographical divisions of China and names of topographies. Elevations are colour-coded. The latitude and longitude ranges and abbreviations of the names of the geographical divisions are located in the centre. The names of China's major terrains are located on the geographical division maps. The size of the terrain font represents the priority of the terrain. The locations of cities are marked by solid dots (red). The bold line (red) in the centre figure is the dividing line between the monsoon and non-monsoon zones. The rectangular boxes are the scale of eight geographical divisions

events. This type of study focuses on the spatiotemporal evolution (He and Zhai, 2018) and causes of the heavy rainfall events (Fang and Yi, 2013; Yuan *et al.*, 2019; Su and Smith, 2021), such as the main circulation patterns and water vapour transport (Wang *et al.*, 2017). In addition, the relationship between heavy rainfall events at daily scales and secondary hazards (Jin *et al.*, 2017; Park *et al.*, 2020; Bai *et al.*, 2021) has also attracted much attention. Heavy rainfall events on subdaily scales are usually referred to as "extreme hourly precipitation." The three main methods for defining extreme hourly precipitation are fixed thresholds, percentile thresholds, and fitted empirical functions. The study of extreme hourly precipitation at the subdaily scale in China is currently focused on the synoptic background (Luo *et al.*, 2016) and climate characteristics (Li *et al.*, 2013) of extreme hourly precipitation (Yuan *et al.*, 2012; Yu *et al.*, 2014). In addition, the impact of urbanization on the intensity, frequency and occurrence areas of extreme hourly precipitation in China has attracted more attention (Liang and Ding, 2017; Li *et al.*, 2017a; Wu *et al.*, 2019).

The purpose of this study is to determine the spatial patterns within the life cycle of heavy rainfall events and the evolutionary trends of heavy rainfall trajectories (HRTs). The HRT is defined in this paper to clarify (a) the annual spatial distribution characteristics and differences in HRTs in subregions and (b) the seasonal spatiotemporal distribution and evolution of the characteristic parameters of HRTs in China.

2 | OVERVIEW OF STUDY AREA AND DATA

2.1 | Study area

Influenced by climate diversity, the characteristics of heavy rainfall in different subregions of China are quite

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TABLE 2 Identification criteria for heavy rainfall events

Rainfall duration (hr)	Cumulative rainfall (mm)	Rainfall intensity ($mm \cdot hr^{-1}$)
≤24	$50 \le q \le 130$	
>24		≥2.0
≤24	$130 < q \le 250$	
>24		≥5.0
≤24	$q > 250$	
>24		≥10.0

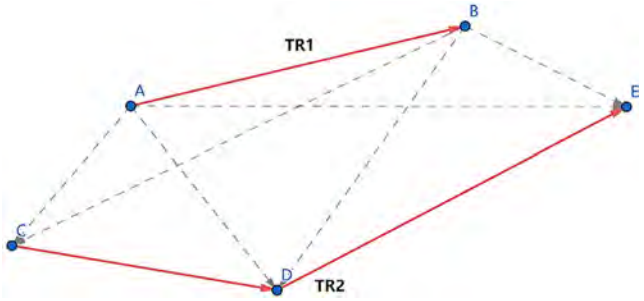


FIGURE 2 The calculation of the dispersion degree

distinct (Luo *et al*., 2020). The differences among the subregions of China regarding the occurrence time of concentrated heavy rainfall events and the types and water vapour sources of heavy rainfall events are summarized in Table 1.

For a detailed understanding of the spatiotemporal distribution patterns of heavy rainfall events in China, this study refers to the geographical divisions in the study of climate patterns in mainland China during summer by Shi and Xu (2006). Eight geographical divisions that result from dividing China (Figure 1): northeast China (NE, east of 110°E, north of 42.5°N); north China (N, east of 110°E, between 35°N and 42.5°N); middle and lower reaches area of Changjiang-Huaihe River (CH, east of 107.5°E, 27.5°–35°N); south China (S, east of 107.5°E, south of 27.5°N); southwest China (SW, 97.5°–107.5°E, south of 35°N); Tibetan Plateau region (TP, west of 97.5°E, south of 35°N); western part of northwest China (NWW, west of 97.5°E, north of 35°N) and Eastern part of northwest China (NEW, 97.5°–110°E, north of 35°N). These eight geographical divisions are indicated by the abbreviations in parentheses in section 4 of this article.

2.2 | Data

To integrate HRTs for hourly-scale heavy rainfall events (heavy rainfall duration <24 hr) and daily scale heavy rainfall events (heavy rainfall duration ≥24 hr) and to investigate the accurate geographic locations of HRTs, this paper uses the highly accurate gridded dataset of hour-by-hour precipitation from the fusion of Chinese automatic stations and the Climate Prediction Center MORPHing technique (CMORPH) precipitation products (version 1.0) published by the National Meteorological Information Center of China Meteorological Administration (CMA, 2021), which is currently included in the real-time product dataset of the China Land Data Assimilation System (CLDAS-V2.0 2021). The selected time range of the data was from the data start time 20080101T000000Z to 20191231T230000Z before the start of the study. The spatial range was 70°–140°E, 15°–60°N with a spatial resolution of 0.1° × 0.1°. Compared with station data, remote sensing rainfall data products have wide coverage and continuous data, which can effectively compensate for the lack of spatial distribution of surface stations.

3 | METHODOLOGY

This paper selects HRTs as an indicator to explore the spatiotemporal distribution and evolutionary trends of heavy rainfall events in China. First, the improved MODE method (Davis *et al*., 2009) and the criterion of heavy rainfall was applied to identify persistent heavy rainfall events. Second, the HRTs were visualized, and the dispersion degree of HRTs (DD) was calculated to describe the spatiotemporal distribution of the HRTs in each season (spring: March–May; summer: June–August; fall: September–November; and winter: December–February in the next year). The DDs, direction angles and lengths of HRTs were quantified, and the evolution of direction angles in the main directions of summer HRTs was investigated.

3.1 | Spatiotemporal distribution characteristics of HRTs

3.1.1 | Identifying heavy rainfall events

This research applies the improved MODE method to identify and extract rainfall events. The MODE method, which is a method of rainfall identification based on object recognition, is utilized by Skok *et al*. (2009, 2013). The algorithm has three main steps. First, the algorithm calculates the convolution of the rainfall data and spatial smoothing operator as the convolution radius (Rs) of the two-dimensional movement; second, disregarding any

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rainfall processes below the threshold (TH); third, the targets are identified as a self-enclosed area of rainfall. We employ the same algorithm, optimal smoothing operator and threshold as those determined by Skok *et al.* (2013). After identifying rainfall events, the heavy rainfall events are identified according to the identification criteria in Table 2 (Zhao and Chen, 2020).

3.1.2 | Spatiotemporal characteristics of heavy rainfall events

This study applies the principle of Fermat points for arbitrary polygons (Ghosh *et al.*, 2009) and the envelopes for scatter points to visualize the HRTs. On this basis, the DD, which is a parameter based on the Euclidean distance, is proposed to quantify the degree of dispersion for HRTs.

Spatial distribution characteristics of HRTs

This study screened the data that formed trajectories and plotted the graphs of HRTs in four seasons for each geographical division. For heavy rainfall events with simultaneous multiple centres, the idea of identifying the Fermat points for arbitrary polygons is applied to equivalently replace the multicentre, which helps to form a distinct direction for HRTs. This method is often applied to the analysis of spatial points and can be expressed as

$$\min\sum d=\sum_{i=1}^{n}\sqrt{(x_0-x_i)^2+(y_0-y_i)^2}. \quad (1)$$

In Equation (1), x_0 and y_0 are the proposed median centres of the point group $\{P\}$, while x_i and y_i are the horizontal coordinate and vertical coordinate, respectively, of point i in the point set $\{P\}$. The sum of the distances from all the points in $\{P\}$ to (x_0,y_0) is $\sum d$.

Distribution characteristics of regional HRTs

The frequency and probability of occurrence of HRTs in a particular spatial location during the study period can be reflected by the dispersion of the HRTs. To quantify the degree of dispersion for HRTs, this study defines a metric parameter based on the Euclidean distance, which is referred to as DD.

The Euclidean distance is a classic method for measuring distance and has been widely applied in data mining and can be employed for path analysis (Yuan *et al.*, 2017). As shown in Figure 2, let {HRT} be the set of trajectories with *n*HRTs, and let $\rightharpoonup$AB and $\rightharpoonup$CDE be heavy rainfall trajectory 1 (HRT1) and heavy rainfall trajectory 2 (HRT2), respectively, in {HRT}. The first step is to calculate the minimum Euclidean distance from point A on HRT1 to all points on HRT2; this distance is defined as $\min\{\|\rightharpoonup AC\|,\|\rightharpoonup AD\|,\|\rightharpoonup AE\|\}$. Second, the distance from the other points on HRT1 to HRT2 can be calculated in the same way. Last, the average of these distances is regarded as the distance from HRT1 to HRT2. Repeating the above process, the distances $d_{1_3},d_{1_4},d_{1_5},\ldots,d_{1_n}$ between HRT1 and the other $n-1$ HRTs in {HRT} can be obtained. Note that the distance between a certain *HRT* and itself is 0.

After the distances $d_{i_1},d_{i_2},d_{i_3},\ldots,d_{i_n}$ are calculated for each HRT to the other HRTs in {HRT}, a $n\times n$

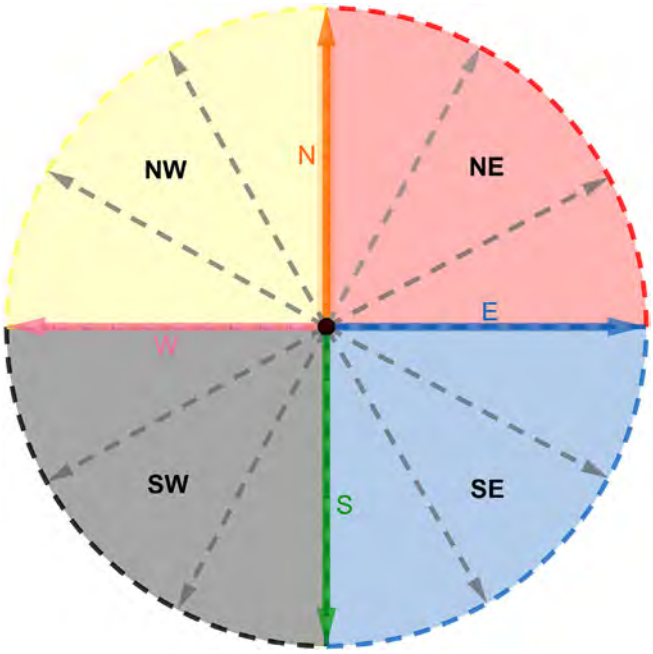


FIGURE 3 Eight directions of the HRTs

TABLE 3 The radius of DBSCAN

Area	Radius
NE	1
N	0.75
CH	0.75
S	0.5
SW	0.75
TP	0.75
NWW	2
NWE	1 (NE* and E*), 1.5 (SE*, SW*, and NW*)
China	0.5

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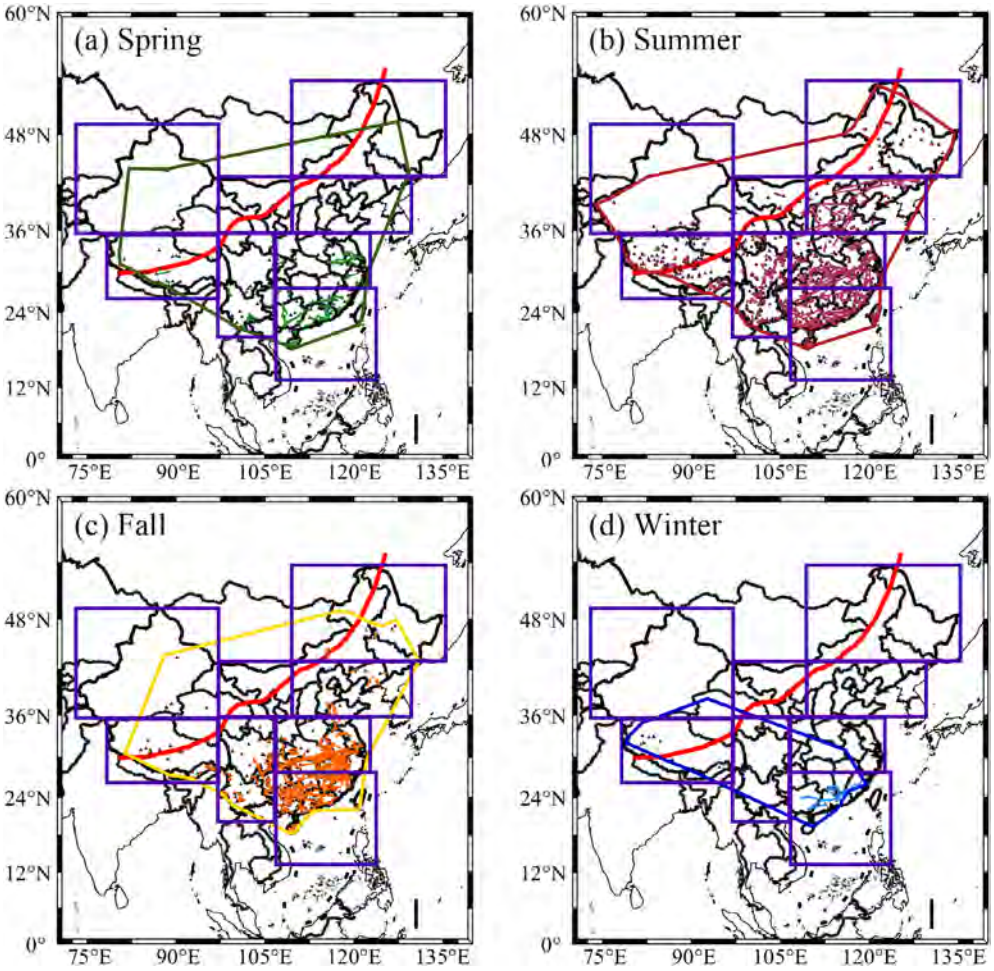


FIGURE 4 Spatial distribution of seasonal HRTs in China in 2016. (a–d) The spatial distributions of HRTs in spring, summer, fall and winter in China, respectively. The triangle symbols and solid dots are the starting and ending points of the HRTs, respectively. The coloured lines are the HRTs and the bold lines are the envelops of heavy rainfall events. The boldest lines are the dividing line between the monsoon and non-monsoon zones. The rectangular boxes are the scale of eight geographical divisions

distance matrix Λ can be obtained. The DD is defined as the average of this distance matrix Λ, with the exception of the diagonal elements, that is,

$$DD = \sum \Psi / NUM(\Psi). \quad (2)$$

In Equation (2), $\sum \Psi$ is the summation of matrix Λ after removing the diagonal elements, and NUM(Ψ) represents the number of matrix Λ after removing the diagonal elements.

The number of HRTs is considered the column of the distance matrix in the process of calculating the DD, so the DD can quantify whether the HRTs spatially concentrate during a season. A larger DD means a greater distance between the HRTs, which means that the spatial area of the HRT distribution is larger in the same region. In contrast, a smaller DD means a smaller distance between the HRTs and that the spatial area of the HRT distribution is smaller in the same region.

3.2 | Characteristics and evolutionary trends of HRTs

The direction of the HRT formed by certain persistent heavy rainfall events is usually related to the direction of water vapour transport, so an investigation of the spatial distribution and evolution of the HRTs is helpful in reflecting the direction and evolutionary trend of water vapour transport. Based on the starting and ending locations of HRTs, the main directional components of the season are selected by the percentage of HRTs in eight directions during a season (Figure 3 and Table 6).

3.2.1 | Spatial distribution characteristics of the DDs, and the length and direction of HRTs

To determine the spatial distribution of the DDs and the length and direction of HRTs in summer China, the starting points of the HRTs during summer in each of the

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TABLE 4 Mean and standard deviation of DD within eight geographical divisions in China from 2008 to 2019

Season	Item	NE	N	CH	S	SW	TP	NWW	NWE
Spring	Mean	5.69	5.98	4.97	4.48	5.00	6.43	6.98	3.38
Summer		6.18	5.20	5.61	5.22	5.32	5.10	6.39	5.00
Fall		6.04	5.19	5.89	5.26	5.09	6.98	7.694	4.42
Winter		0.02	4.12	3.78	4.69	4.87	5.02	7.52	5.60
Spring	Standard deviation	1.34	4.34	0.49	0.22	0.57	0.85	3.70	1.53
Summer		0.55	0.50	0.36	0.35	0.31	0.44	1.43	0.64
Fall		1.90	1.62	0.76	0.64	0.51	2.01	3.15	3.20
Winter		0.01	0	1.58	1.06	2.04	1.40	3.17	0

Note: The mean of DD is the interannual average of the dispersion degree of HRTs in a certain season in a certain region from 2008 to 2019, which can be used to compare the differences in dispersion degree of HRTs in different geographical divisions within the same season or the differences in dispersion degree of HRTs in one region in different seasons. The standard deviation of DD is the fluctuation of DD in a region in selected season over the years 2008–2019. The standard deviation of DD allows us to analyse whether the fluctuation of DD is stable in a region within a season.

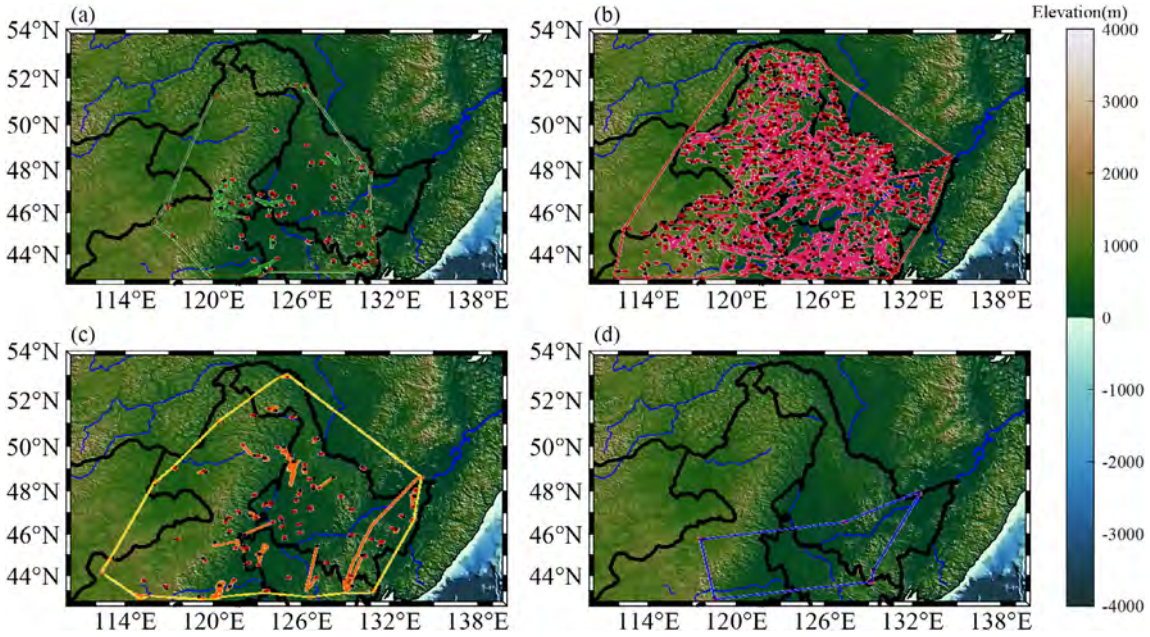


FIGURE 5 Spatial distribution of seasonal HRTs in NE during the study period. (a–d) The spatial distributions of HRTs in spring, summer, fall and winter in NE, respectively. The deep dots (black) and light dots (red) are the starting and ending points of the HRTs, respectively

geographical divisions are clustered by density-based spatial clustering of applications with noise (DBSCAN) to determine the representative classes. The minimum number of DBSCAN is set to 3, and the radiuses of DBSCAN set in this research are shown in Table 3. The DDs for the major directions in summer and the lengths and direction angles of the seasonal directional vectors are then calculated for every class. The centres of every class are calculated by the Fermat points for arbitrary polygons (mentioned in section 3.1.2) to establish the spatial distributions of these three parameters.

3.2.2 | Spatiotemporal evolution of the direction of HRTs

EOF analysis is performed to investigate the spatiotemporal evolution of the major direction angles. EOF, a spatiotemporal analysis method, is widely utilized in meteorological and climate studies to analyse the structural features in matrix data to extract the eigenvectors (Alexander *et al.*, 2015). The purpose of this method is to represent the spatial and temporal characteristics of the physical fields by obtaining eigenvectors and time coefficients. Compared

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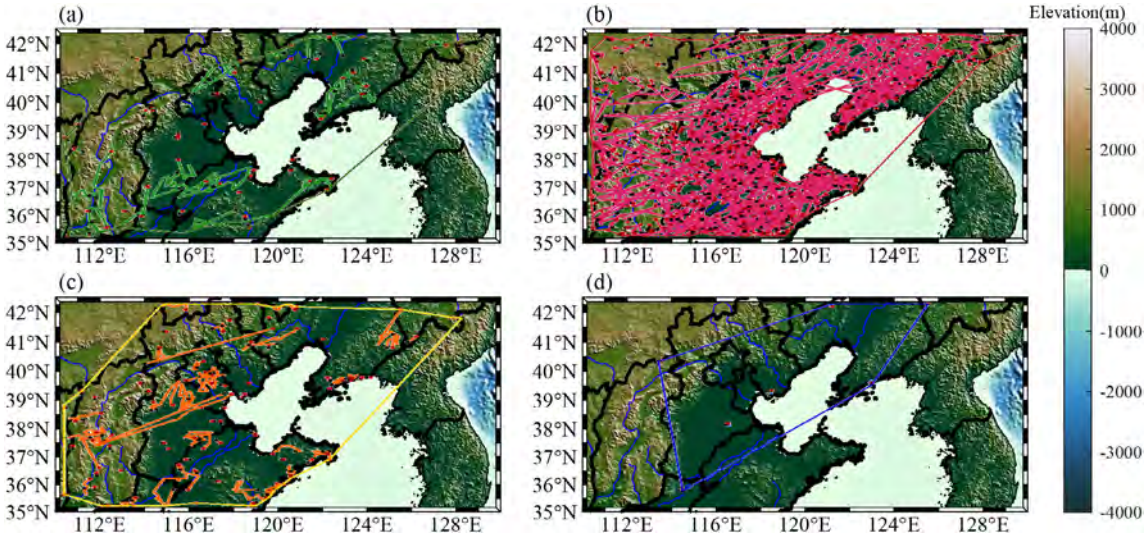


FIGURE 6 Spatial distribution of seasonal HRTs in N during the study period. (a–d) The spatial distribution of HRTs in spring, summer, fall and winter in N, respectively. The deep dots (black) and light dots (red) are the starting and ending points of the HRTs, respectively

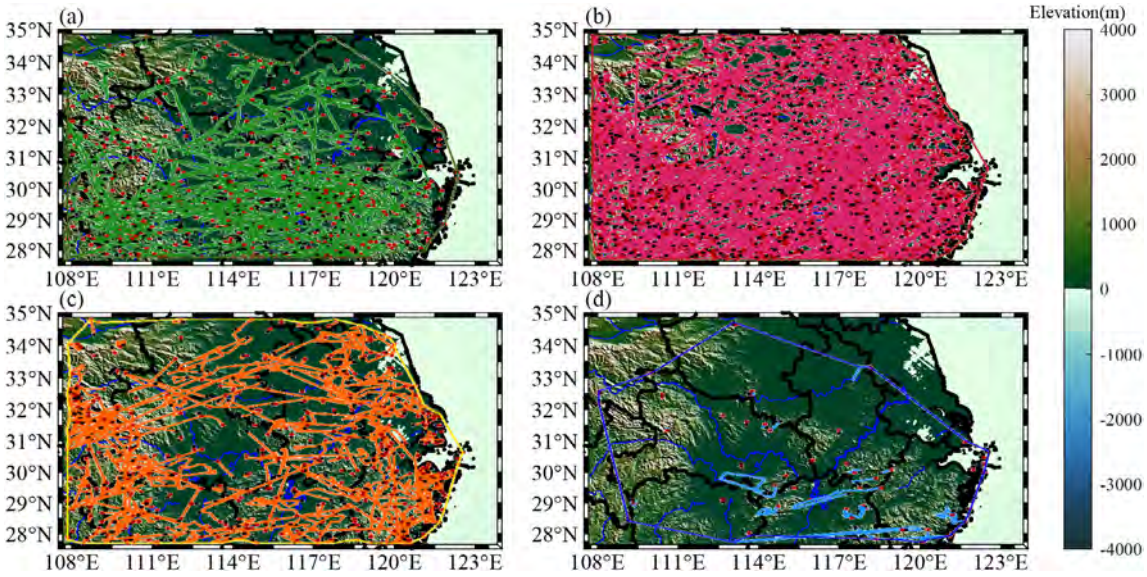


FIGURE 7 Spatial distribution of seasonal HRTs in CH during the study period. (a–d) The spatial distribution of HRTs in spring, summer, fall and winter in CH, respectively. The deep dots (black) and light dots (red) are the starting and ending points of the HRTs, respectively

with the last part, DBSCAN is performed in the whole area to obtain the classes of direction angles rather than in the subregions. The minimum number and radius are set to 3 and 0.5, respectively.

To avoid confusion with the abbreviations of the eight geographical divisions, the abbreviations of the eight directions are marked with *. For example, NE* represents the northeast direction, while NE represents northeast China.

4 | RESULTS AND DISCUSSION

The seasonal HRTs of China for each year are visualized. These HRTs have distinctive spatial and seasonal characteristics (Luo *et al.*, 2020). The spatial distribution of the HRTs in 2016 is typical and can represent the most usual distribution of HRTs in the four seasons (Figure 4). First, summer is the season with the most and concentrated HRTs in China; second, there are more HRTs in eastern

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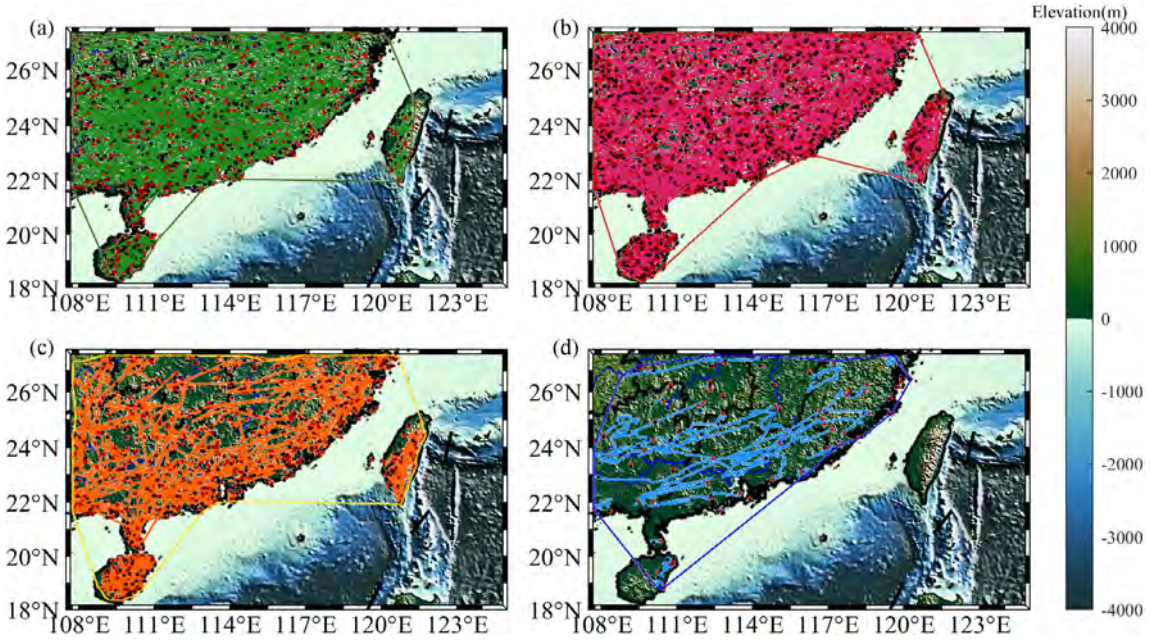


FIGURE 8 Spatial distribution of seasonal HRTs in S during the study period. (a–d) The spatial distribution of HRTs in spring, summer, fall and winter in S, respectively. The deep dots (black) and light dots (red) are the starting and ending points of the HRTs, respectively

China than in western China in every season; third, HRTs in eastern China are longer than those in western China. To obtain more accurate results, the following section is designed to learn the characteristics of HRTs in eight subregions.

4.1 | Intra-annual spatiotemporal distribution of HRTs within eight geographical divisions of China

The sources of water vapour that contribute to heavy rainfall events (Liu *et al.*, 2018) and the frequency of water vapour transport that occurs are two important factors of the DD. A smaller DD indicates that (a) heavy rainfall events are frequent within a season with relatively homogeneous sources of water vapour and less interannual variation in the distribution of moisture channels (standard deviation of DD < 1); (b) heavy rainfall events are infrequent within a season (the number of interannual heavy rainfall events within this season from 2008 to 2019 < 2020); and (c) heavy rainfall events are infrequent within a season (the number of interannual heavy rainfall events within this season from 2008 to 2019 > 2020) but are significantly influenced by topography.

In this section, the areas of concentrated HRTs in four seasons within eight geographical divisions and the possible causes of the magnitudes of the DD values within each season are analysed. Table 4 shows the mean and standard deviation of DD within eight geographical divisions in China. Figures 5–12 show the visualization results of the four-season HRTs within the eight geographical divisions. The areas of concentrated HRTs are obtained from Figure 1 and the visualization results in Figures 5–12; they are listed in Table 5.

The blocking effect of mountains on the HRTs is very obvious in spring. In coastal areas, heavy rainfall events show the characteristics of long trajectories and wide extensions. The HRTs in NE and in N are mainly concentrated in the plains; in the NE, some of the HRTs follow the tributary valleys of the upper reaches of the Songhua River into the western side of the Da Hinggan Mountains, forming an area of concentrated HRTs in the valley areas of the Inner Mongolia Plateau (Figure 5a); in the N, Mount Taishan prevents the HRTs from moving further into northeast Shandong Province (Figure 6a), and the small-scale area of concentrated HRTs formed in central Shanxi (Taiyuan) is contributed by persistent heavy rainfall events and prevented from moving northward by the Taihang Mountains (Kang *et al.*, 2020). Note that in Table 4, the standard deviation in spring of 4.34 (>1) in the N is much higher than that of other geographical divisions in spring. This finding shows that the N in spring has the weakest regularity of heavy rainfall events, which may be related to the complex causes of heavy rainfall events in N in spring (Luo *et al.*, 2020). Persistent

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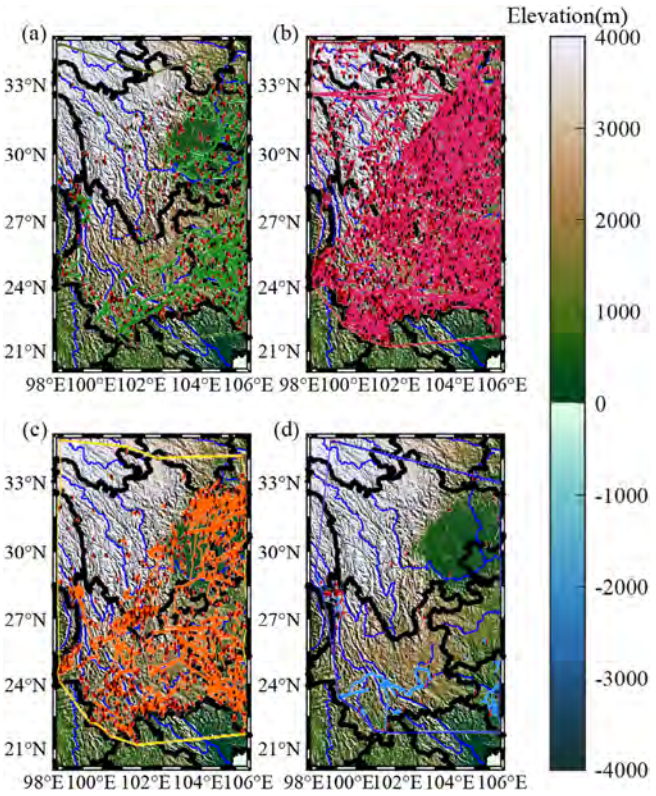


FIGURE 9 Spatial distribution of seasonal HRTs in SW during the study period. (a–d) The spatial distribution of HRTs in spring, summer, fall and winter in SW, respectively. The deep dots (black) and light dots (red) are the starting and ending points of the HRTs, respectively

heavy rainfall events in northwest China (NWW and NWE) are mostly concentrated along mountain ranges with minimal regional variation in the distribution during the year (Figures 11a and 12a).

The stability of the summer DDs is the strongest, which indicates that the HRTs in summer are more regular in space than those in other seasons. The areas of concentrated HRTs are expanded with longer and more extended HRTs, while the blocking effect of mountains on HRTs considerably varies within different geographical divisions. In NE, the areas of concentrated HRTs in summer extend to the entire region with longer and more extensive HRTs, but the HRTs are distinctly separated by mountains (Figure 5b). In N, the HRTs are generally blocked by the Taihang Mountains, showing a pattern of more HRTs on the eastern side and fewer HRTs on the western side, while some HRTs can develop across the Taihang Mountains. However, the blocking effect of Mount Taishan on the HRTs is not obvious (Figure 6b). When the Meiyu Period on the Changjiang-Huaihe River is in progress, the blocking effect of the mountains on the HRTs in CH is not obvious in the coastal areas (Figure 7b). The distribution area of HRTs in S is larger than that in spring, and the blocking effect of mountains is not significant in summer (Figure 8b). The two areas of concentrated HRTs in SW extend to the eastern side of the Hengduan Mountains in summer, which is the maximum area of concentrated HRTs throughout the year in this region (Figure 9b). The distribution characteristics of the HRTs in summer and spring on the TP are consistent, and persistent heavy rainfall events in summer are frequent and regular (4-season standard deviations are less than 1.40) in northwest China (NWW and NWE) (Figures 11b and 12b).

There is a strong similarity in the spatial distribution of HRTs between fall and spring. The number of persistent heavy rainfall events in the fall is plummeting, and the length of HRTs is decreasing. The HRTs in NE and in N are also mainly located in the plains, but the HRTs are distributed over a wider range of plains than those in spring, such as those in the Sanjiang Plain; in addition, the area of concentrated HRTs located in Taiyuan still exists, while there appears to be a novel area of concentrated HRTs in the coastal area of Shandong Province (Figures 5c and 6c). In the fall, summer monsoons move southward to CH. Under the influence of the western Pacific subtropical high and anticyclones over the Philippine Islands (Qian *et al.*, 2021), heavy rainfall events in coastal areas and the western part of CH extend along the mountain ranges to the middle and lower reaches of the plain of the Changjiang River (Figure 7c). In SW, the areas of concentrated HRTs in the fall are similar to but less extensive than those in the summer (Figure 9c). On the TP, the HRTs located in the southern part of the northern Tibetan Plateau are shorter than those in the southeastern Tibetan Plateau (Figure 10c).

Very few persistent heavy rainfall events occur during winter in China. The spatial distribution of HRTs shows the characteristics of more HRTs in the southern part of China than in the northern part of China, and they are almost evenly distributed in plains. The heavy rainfall events in NE are low in number (less than five HRTs in 2013, 2015 and 2016) and spatially concentrated in Jilin (especially concentrated around Songhua Lake) (Figure 5d), which may explain the low value of the DD in winter. With short HRTs, the winter heavy rainfall events in CH are mostly blocked by mountains, so they are distributed on the southern side of Dabie Mountain (Figure 7d). In SW, the HRTs are shorter on the southwestern side of the Hengduan Mountains than on the southeastern side of the Yunnan-Guizhou Plateau (Figure 9d). On the TP, the spatial distribution characteristics of HRTs in winter and fall are consistent, but the HRTs located in the southeastern Tibetan Plateau are significantly discrete (Figure 10d).

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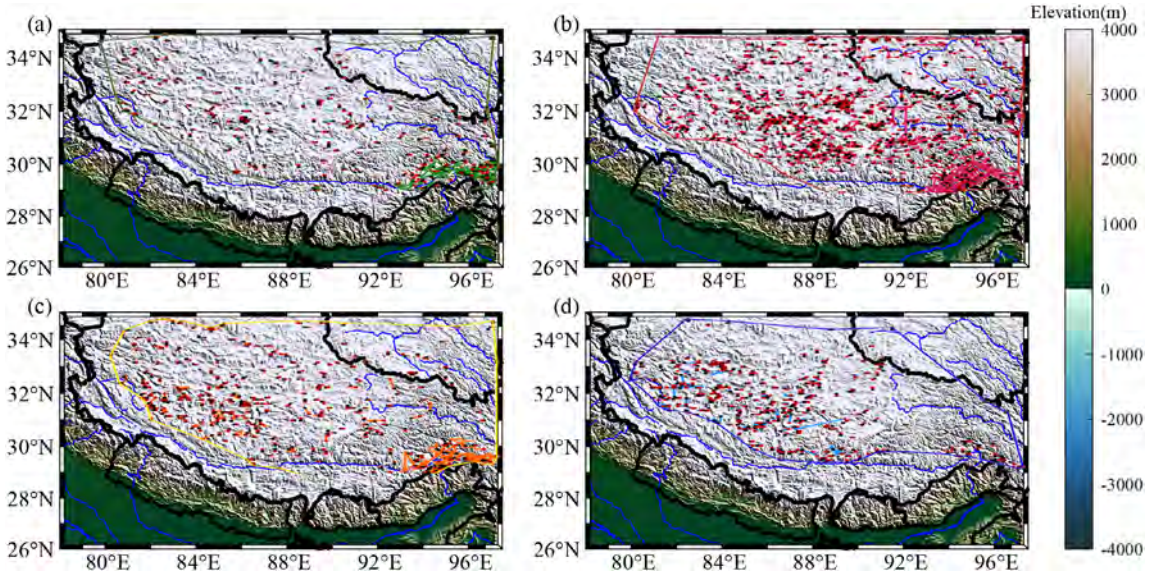


FIGURE 10 Spatial distribution of seasonal HRTs in TP during the study period. (a–d) The spatial distribution of HRTs in spring, summer, fall and winter in TP, respectively. The deep dots (black) and light dots (red) are the starting and ending points of the HRTs, respectively

Persistent heavy rainfall events often cause flooding and other disasters (Jiang *et al*., 2013). Table 5 shows that the spatial distribution areas of persistent heavy rainfall events in SW and NWE are more fixed in a year, so there will be more severe secondary hazards in the areas of concentrated HRTs. Many studies have demonstrated the close relationship between heavy rainfall events and the Taihang Mountains in N. The Taihang Mountains can effectively trigger and amplify mesoscale convective systems, leading to a concentration of heavy rainfall events on the eastern side of the mountains (Sun, 2005; Kang *et al*., 2020), which is generally consistent with the results in this study for the area of concentrated HRTs in N. In addition, note that in N the areas of concentrated HRTs are formed in Beijing during both summer and fall, which has the same temporal distribution of intense hourly rainfall events in the warm season as obtained by Li *et al*. (2017b). In Taiwan, the areas of concentrated HRTs occur on the western side of the Central Range in spring and on the eastern side in fall, which is also corroborated by the seasonal study of extreme rainfalls in Taiwan by Henny *et al*. (2021).

The HRTs in CH and in S have the strongest regularities within every season (standard deviation less than 1.5). The interseasonal range of DD in CH is greater than that in S, which may be attributed to differences in the duration of the East Asian monsoon and water vapour sources. S is the subregion with the longest period affected by the East Asian summer monsoon in a year; on the other hand, the water vapour sources that produce heavy rainfall events in CH are complex (Huang *et al*., 2012), which exacerbates the spatially disordered distribution of HRTs. Two elements contribute to the annually stable DDs in SW. First, the weather systems that affect heavy rainfall events in SW are relatively homogeneous (southwest China vortex) (Luo *et al*., 2020), although heavy rainfall events mainly occur in summer. Second, the number of heavy rainfall events in other seasons is small, with a concentrated distribution area (Figure 9). The relationship between mountains and heavy rainfall events is noteworthy. In northwest China, this work revealed that the Tianshan Mountains and Kunlun Mountains can block HRTs and lift water vapour to produce heavy rainfall events.

4.2 | Spatiotemporal distribution and evolution of HRT parameters in summer

The analysis in section 4.1 indicates that persistent heavy rainfall events in China are frequent with strong regularity and predictability in summer. To further study the variability in the directional components of HRTs for different geographical divisions and their interannual variation patterns, the spatiotemporal distribution and evolution of the summer HRTs were analysed by clustering and EOF analysis.

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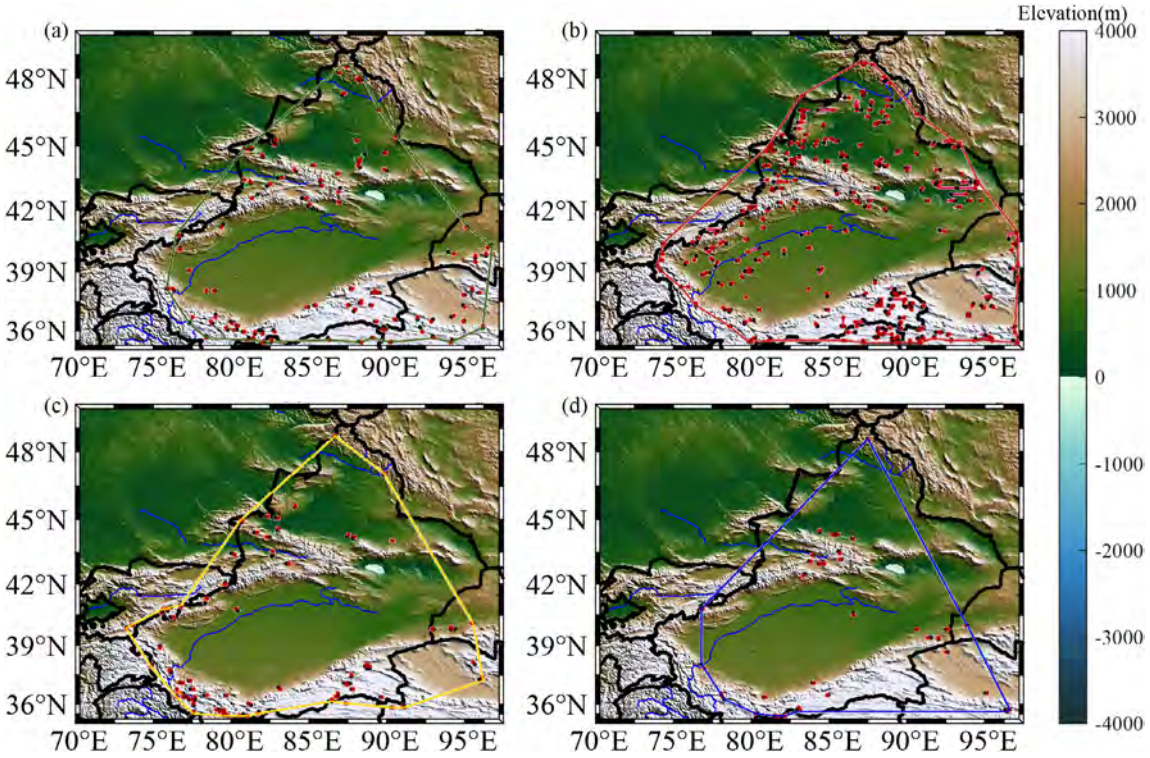


FIGURE 11 Spatial distribution of seasonal HRTs in NWW during the study period. (a–d) The spatial distribution of HRTs in spring, summer, fall and winter in NWW, respectively. The deep dots (black) and light dots (red) are the starting and ending points of the HRTs, respectively

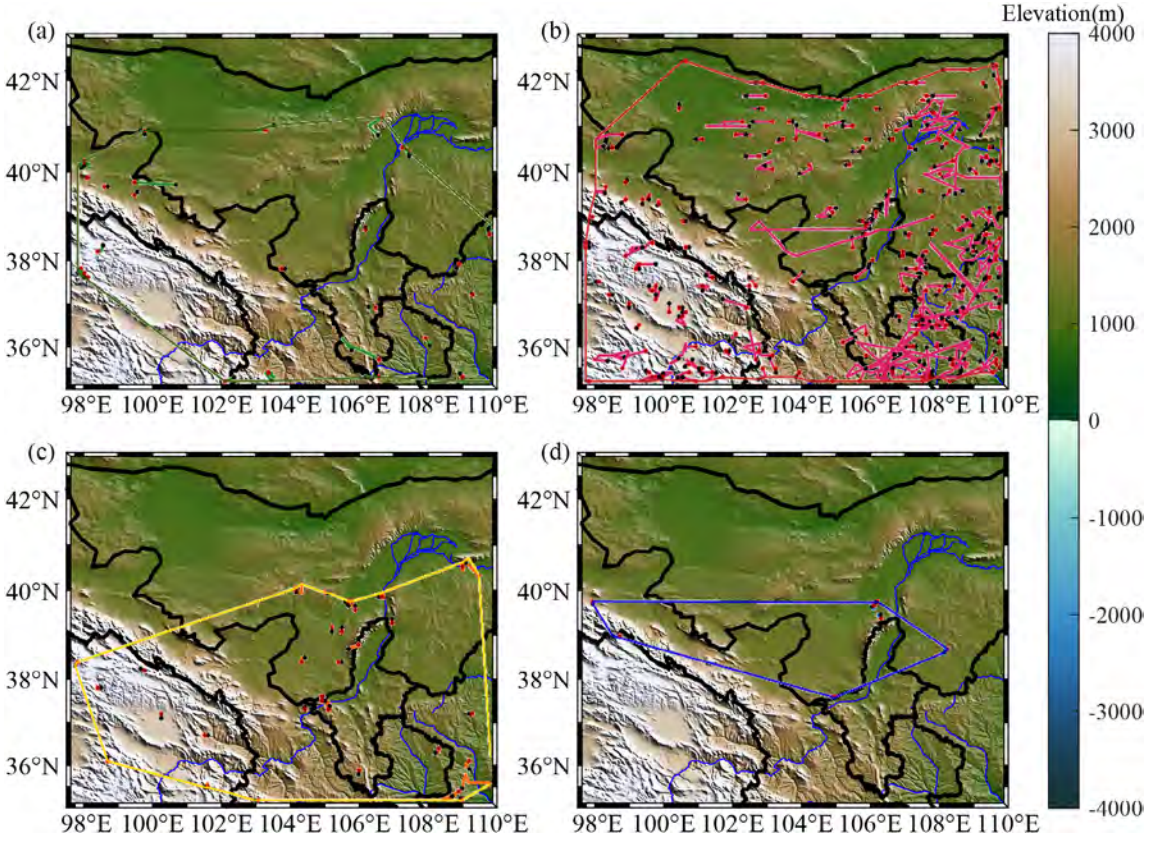


FIGURE 12 Spatial distribution of seasonal HRTs in NWE during the study period. (a–d) The spatial distribution of HRTs in spring, summer, fall and winter in NWE, respectively. The deep dots (black) and light dots (red) are the starting and ending points of the HRTs, respectively

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TABLE 5 The areas of concentrated HRTs in four seasons within eight geographical divisions

Region of China	Spring (Mar–May)	Summer (Jun–Aug)	Fall (Sep–Nov)	Winter (Dec–Feb)
NE	Songnen Plain River valley area in Inner-Mongolia Plateau	Songnen Plain Liaohe Plain	Sanjiang Plain Northeast China Plain	Jilin (especially Songhua Lake)
N	The Yellow River Basin in North China Plain Taiyuan	Bohai Gulf Central Shanxi Taiyuan	The Haihe River Basin in North China Plain Taiyuan Beijing	
CH	Middle and Lower Reaches Plain of Changjiang River Jiangnan Hills in the southern side of Dabie Mountains	The eastern side of Qinling Mountains and southern side of Dabie Mountains	Coastal areas The plain in the southern side of Daba Mountain	The mountainous areas in southern side of Wuhan and eastern side of Changsha
S	Southern side of Yunnan-Guizhou Plateau Zhujiang Delta Hainan Island Western side of the central range	Area except the Nanling Mountains	Coastal areas Hainan Island Eastern side of the central range	Zhujiang Delta
SW	Southeastern side of Yunnan-Guizhou Plateau Sichuan Basin			Southwestern side of Hengduan Mountains Southeastern side of Yunnan-Guizhou Plateau
TP	Southeast Tibetan Plateau	Southeast Tibetan Plateau Southern part of Northern Tibetan Plateau	Southern side of northern Tibetan Plateau Southeast Tibetan Plateau	Southeast Tibetan Plateau
NWW	Tianshan Mountains Kunlun Mountains Eastern side of Qilian Mountain Southern side of Altay Mountains	Junggar Basin Tarim Basin	Tianshan Mountains Kunlun Mountains	Tianshan Mountains
NWE	Yellow River Plain in the southern side of Yinshan Mountain Loess Plateau	Loess Plateau Southern side of Yinshan Mountain	Helan Mountain	

TABLE 6 Statistics on the direction of HRTs from 2008 to 2019 (%)

Region of China	Start and end points overlap	N*	NE*	E*	SE*	S*	SW*	W*	NW*
NE	0.13	2.88	23.01	17.258	17.78	3.53	11.37	9.15	14.90
N	0	2.97	30.88	8.14	18.48	2.33	16.54	5.30	15.37
CH	0.06	4.39	29.08	9.51	19.02	2.83	15.29	5.42	14.39
S	0	5.62	29.048	8.12	13.61	4.02	15.00	5.66	18.93
SW	0	5.52	25.74	6.79	18.53	4.45	16.61	5.15	17.22
TP	0.20	8.25	18.47	21.41	11.69	5.40	13.65	10.90	10.02
NWW	0.32	4.55	12.01	32.47	10.06	4.22	9.42	17.86	9.09
NWE	0	3.67	27.52	15.14	19.27	3.67	13.76	8.26	8.72

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TABLE 7 Correlation of DDs, length and direction angle of HRTs

Parameters	Direction angle	DDs	Trajectory length
Direction angle	1	0.13	−0.17
DDs	0.13	1	−0.16
Trajectory length	−0.17	−0.16	1

Note: The DD, trajectory length and direction angle of HRTs are both weakly correlated.

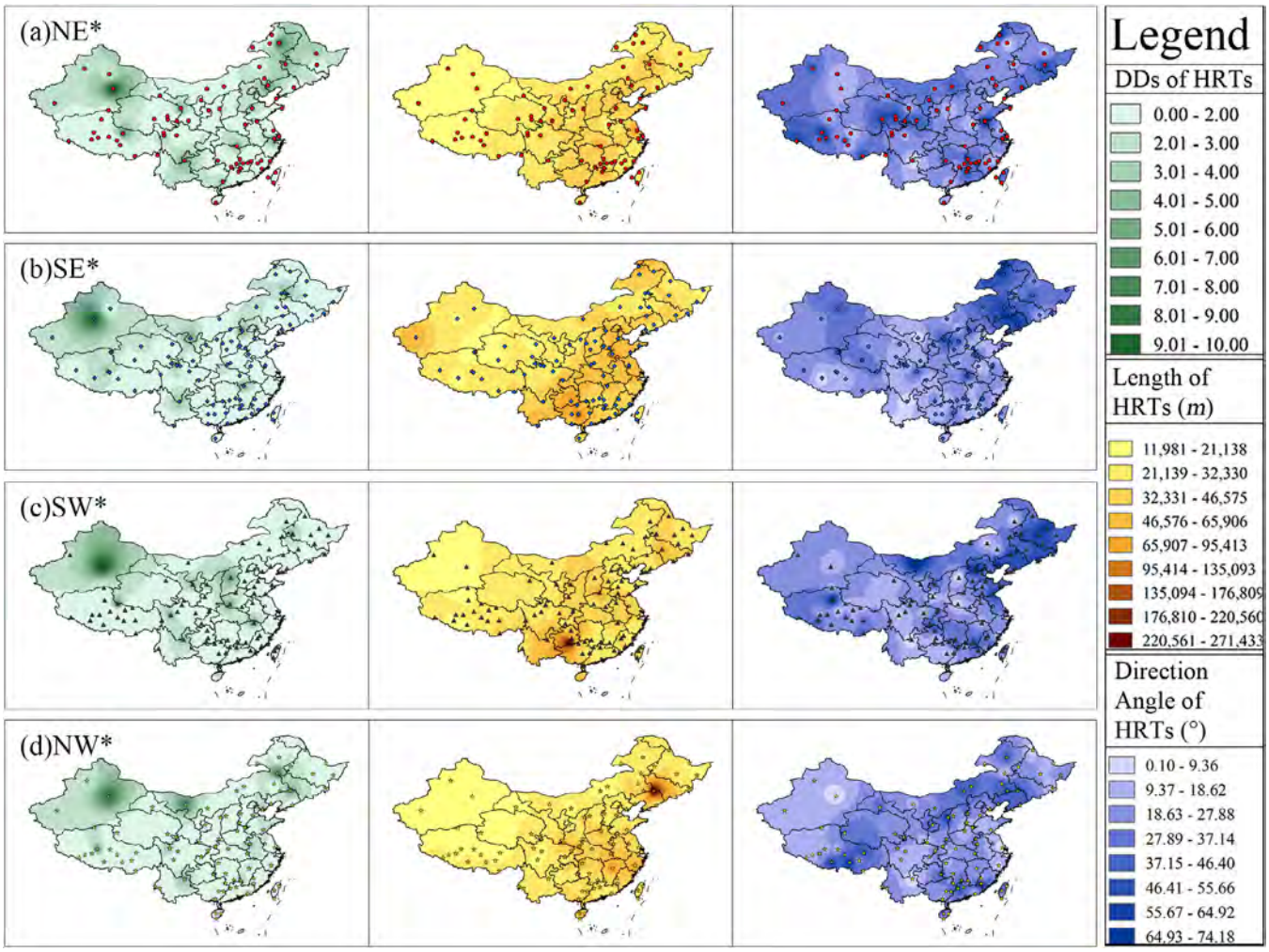


FIGURE 13 Spatial distribution of the DDs, lengths and direction angles of summer HRTs in four main directions. Each row represents the DDs, lengths and direction angles of summer HRTs in one direction, and the first to third columns represent the DDs, lengths and direction angles for the four main directions, respectively. The filled contours in the first, second and third row are the value ranges of DDs, lengths and direction angles of summer HRTs, respectively. The solid dots (red), diamonds (blue), triangles (grey) and stars (yellow) are the clustering centres of the starting points of summer HRTs by DBSCAN in the NE*, SE*, SW* and NW* directions, respectively. The units of length and direction angles of HRTs are meters (m) and degrees (°). The real values of the direction angles of HRTs in SE*, SW* and NW* need to be calculated as the method of adding 90, 180 and 270 to the values in the legend

4.2.1 | Directional components and spatiotemporal distribution of HRTs parameters

According to the data in Table 6, persistent heavy rainfall events in summer tend to move eastward (NE*, SE* and E*). The main directions of HRTs in summer are northeast (12–30%), southeast (10–20%) and southwest (9–16%). With the exception of northwest China (NWW and NWE), the percentages of northwest-oriented HRTs are also high (10–19%). The high percentages of directly east-oriented HRTs appear in northwest China (NWW and NWE), the TP and NE (15–32%). In addition, west-

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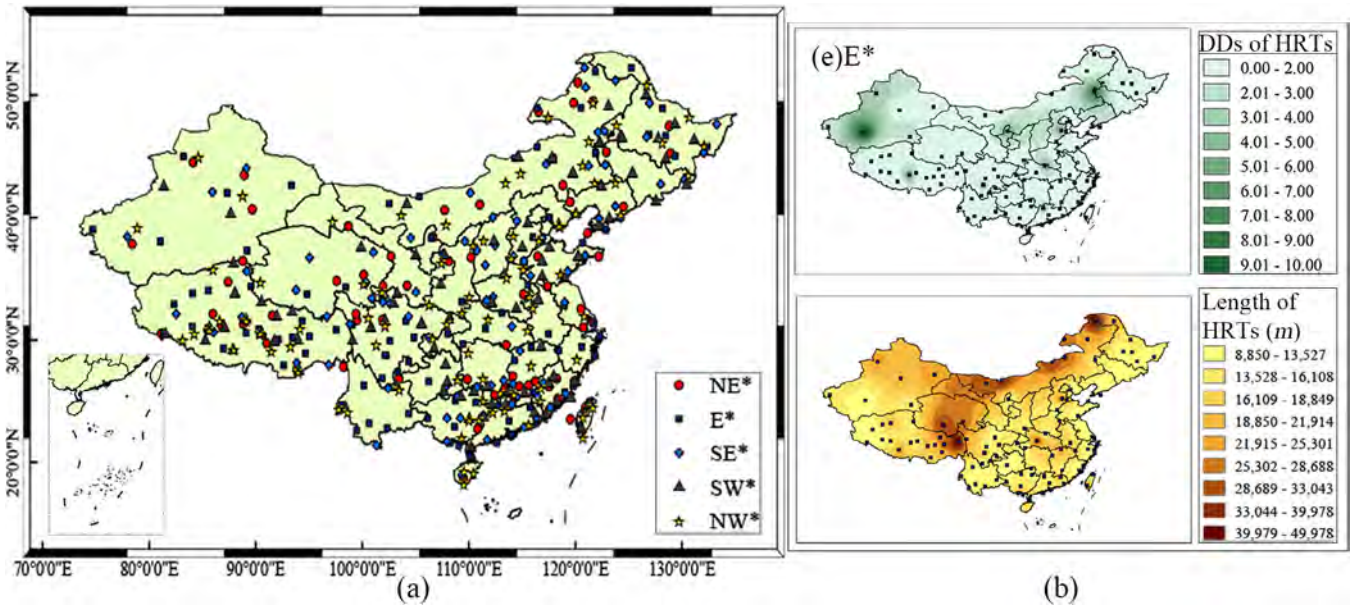


FIGURE 14 The distribution of clustering centres and spatial distribution of the DDs, lengths of summer HRTs in E*. (a) The left part shows the distribution of clustering centres in the main summer directions. (b) The right part is the spatial distribution of the DDs, lengths of summer HRTs in E*. The filled contours in the upper and lower figure are the value ranges of the DDs, lengths of summer HRTs, respectively. The squares (deep blue) are the clustering centres of the starting points of summer HRTs by DBSCAN in the E* direction. The length of HRTs is measured in meter (m)

TABLE 8 Correlation of DD, length and direction angle with direction of HRTs

Parameters	Direction angle	DD	Length
Direction	0.98	0.08	−0.31

oriented HRTs are very common in NWW and in TP (10.90 and 17.86%, respectively).

The northeast, directly east, southeast, southwest and northwest directions of HRTs account for more than 70% of the total directional components in eight geographic divisions. Thus, these five directions of HRTs were selected as the main objects in this study to analyse their spatial distribution pattern of DDs, length and direction angle in summer. Since the direction angle directly east is constant (90°), only the direction angles of the northeast, southeast, southwest and northwest in summer are analysed in the direction angle study.

The class centre of each direction is displayed in Figure 14a. Nationwide, there is a weak negative correlation of space between the DDs and the length of HRTs (Table 7). From southeast China to northwest China, the DDs increase to 8 and the length decreases by 53,925 m (Figure 13). The northeastward HRTs in NE, N, NWW and SW disperse with DD values above 5, while the lengths in these four subregions rank as N (32,331–46,575 m), NE, SW (21,139–32,330 m) and NWW (11,981–21,138 m). However, in CH, S, TP and NWE, all the DDs are below 5, and the lengths of HRTs are consistent with the pattern that HRTs in the east are longer than those in the west. The distribution of direction angles seems more complex, but a low-value area can be observed alongside the dividing line between the monsoon zone and the non-monsoon zone, which means that the HRTs along the dividing line are steeper than those in other places (Figure 13a).

High-value lengths for southward HRTs (SE* and SW*) can be obtained in the monsoon zone, particularly on the southeastern side of Yunnan-Guizhou Plateau (>65,907 and >176,810 m). In addition, the largest (smallest) direction angles of SE* and SW* are shown in NE (SW), which shows that the southeastward HRTs are steep in NE and that the southwestward HRTs are steep in SW. High-value DDs (>5) of southeast-oriented HRTs are located in NE and NWW, whereas southwestward HRTs (DDs > 5) appear to be more dispersed in NWW, NWE and CH (Figure 13b,c).

Unlike the northeastward HRTs, the dispersed HRTs are distributed in NE, NWW and NWE (>6), and there appears to be an obvious high-value centre of HRT length in NE (>176,810 m). However, the low-value area alongside the dividing line between the monsoon zone and the non-monsoon zone also exists, which means that the HRTs in this area are gentler than those in other places (Figure 13d).

Directly eastward HRTs were also dispersed in the northern part of China (NE, N, NWW and NWE). These

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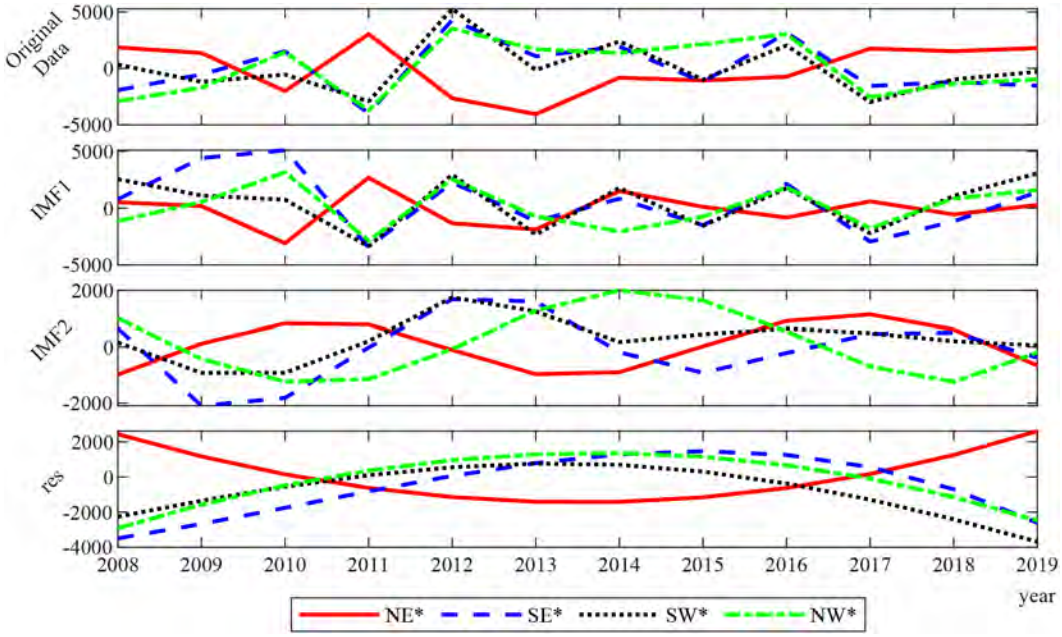


FIGURE 15 The variation form of time coefficients in four main directions. The upper figure shows the original data of corresponding time coefficients. By EEMD, the original time sequences are separated into two intrinsic mode functions (imf). The res is signal residual quantity. The solid lines (red), dashed lines (blue), dotted lines (black) and dash-dotted lines (green) represent the time sequences of NE*, SE*, SW* and NW*, respectively

HRTs are, however, shorter than those in the other directions, but there is a distinct high-value belt (>33,044 m) of HRT length extending from the Xiao Hinggan Mountains to the Tibetan Plateau, which is almost on the north side of the dividing line between the monsoon zone and the non-monsoon zone (Figure 14b).

4.2.2 | Spatiotemporal evolution of the main directions of HRTs

The direction angles of HRTs determine the direction of the HRTs. The DDs do not directly correlate with the direction of the HRTs, and the lengths of the HRTs show negative moderate correlations with the direction of the HRTs (Table 8). Therefore, EOF analysis was performed only for the direction angles of the northeast, southeast, southwest and northwest directions in summer. The variance contribution rates of the eigenvalues for the first eigenvectors in the northeast, southeast, southwest and northwest directions was substantially greater than in other modes. Thus, the first eigenvectors of the directional components in summer can well represent the spatiotemporal evolution type of the direction angles of the HRTs in China between 2008 and 2019. The decomposition of the time coefficients with ensemble empirical mode decomposition (EEMD) (Bi *et al.*, 2020) was utilized to obtain the variation form of the main directions (Figure 15).

The clustering centres of each direction class are displayed in Figure 16a. The spatial evolutionary mode of northeastward HRTs shows a significant negative trend nationwide, which indicates that there is a high consistency in the changing trend of direction angle in summer between 2008 and 2019. Combined with the time coefficients of the EOF decomposition for the direction angles, the northeastward HRTs are the gentlest in 2016 in negative trend areas. From 2016 to 2019, the direction angles of the northeastward HRTs continued to slowly decrease, but they did not become as steep as observed in 2008.

The changing trends of the southeastward, southwestward and northwestward HRTs show a significant positive trend nationwide. In addition, their time coefficients show the same changing trend after 2015. In the positive trend areas, the southeastward HRTs are the steepest in 2016, while they become increasingly gentler after 2016. The changing trend of the direction angles of the southwestward HRTs has three obvious stages, which serve as the split points of these stages in 2011 and 2017. In the positive trend areas, the southwestward HRTs become increasingly steeper before 2011 but become gentler until 2017 before their direction angles slowly decrease. The changing trend in positive areas of northwestward HRTs is similar to that of southeastward HRTs, while the increasing range of the direction angles of northwestward HRTs is larger from 2012 to 2016.

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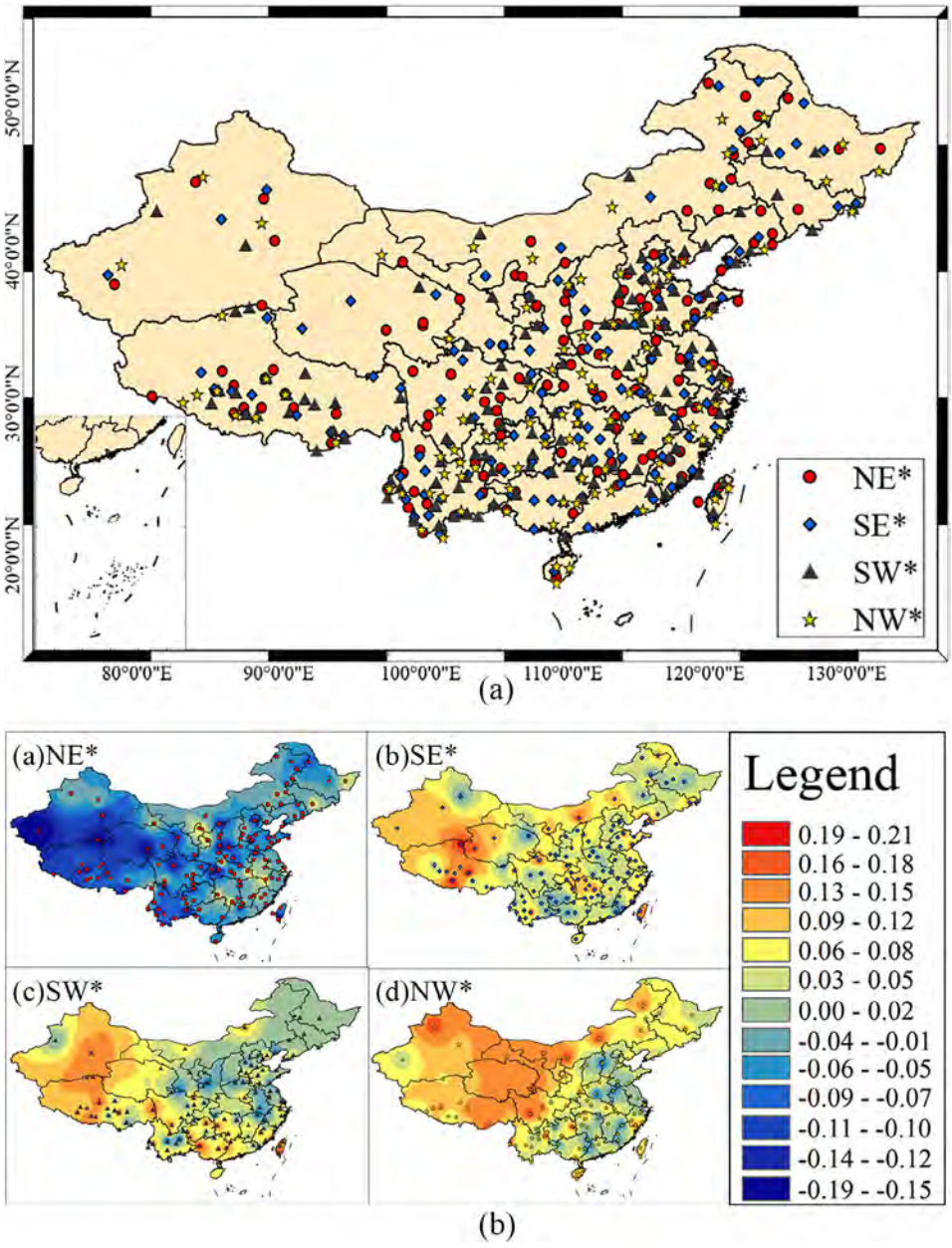


FIGURE 16 The distribution of clustering centres and direction angle evolution mode of four main directions in summer. (a) The distribution of clustering centres in the four summer directions (dots: NE*; diamonds: SE*; triangles: SW*; stars: NW*). (b) The direction angle evolution mode of the four main directions in summer. The filled contours are the value ranges of the magnitude of the change in direction angles of HRTs in summer

From 2008 to 2019, the northeast-oriented HRTs slowed, and the HRTs in eastern China were flatter than those in western China, which showed a southward deflection from west to east. The HRTs in the southeast direction steepened, with steeper HRTs in western China than in eastern China, which showed a northward deflection from west to east. From 2008 to 2019, the southwestward HRTs slowed, and the HRTs in western China were flatter than those in the eastern China, which showed a northward deflection from east to west. The northwestward HRTs steepened, and the HRTs in western China were steeper than those in eastern China, which showed a northward deflection from east to west.

5 | CONCLUSION

This paper investigates the spatiotemporal patterns of HRTs in China based on the time-by-time rainfall 0.1° grid dataset (version 1.0) fused with the COMORPH precipitation product. We draw the following conclusions.

1. The spatial distribution and length of HRTs in China have distinct geographical and seasonal characteristics, and the sequence of persistent heavy rainfall events occurring in eight geographical divisions has a strong consistency with the summer monsoons' northward advance and southward retreat. In addition, the blocking and facilitative effects of mountains on heavy

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rainfall events are corroborated from the perspective of HRTs.

2. From southeast China to northwest China, the DDs increased to 8 and the length decreased by 53,925 m. The northward (northwestward) HRTs are the steepest (gentlest) along the dividing line between the monsoon zone and non-monsoon zone. The southward HRTs (SE* & SW*) are the longest on the southeastern side of the Yunnan-Guizhou Plateau (>65,907 and >176,810 m).
3. Across the country, the direction angle changing patterns of northeast-oriented HRTs show a significant negative trend, whereas southeast-, southwest- and northwest-oriented HRTs show a significant positive trend. From 2008 to 2019, HRTs flattened in the northeast and southwest directions but were steeper in the southeast and northwest directions.

In our next study, we will use a database with a longer time series to learn the long-series spatiotemporal evolution of heavy rainfall events. In addition, quantifying the complete shape of the HRTs will be the highlight of the next study.

AUTHOR CONTRIBUTIONS

Chang Xu: Data curation; formal analysis; methodology; software; visualization; writing – original draft; writing – review and editing. **Zhihe Chen**: Conceptualization; funding acquisition; resources; supervision; writing – review and editing. **Zhihua Zhu**: Data curation; funding acquisition.

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